



Electric Vehicles and Charging Infrastructure in Sustainable Transport Systems: Technologies, Integration Challenges, and Optimisation Pathways

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Abstract: Electric vehicle (EV) technologies and charging infrastructure have developed rapidly, placing increasing pressure on transport systems to accommodate new forms of energy demand and mobility. While substantial progress has been made in individual technologies, system-level questions—particularly those related to infrastructure integration, interoperability, and coordination with energy networks—remain insufficiently addressed. This study provides a structured review of EV charging technologies and associated optimisation approaches from a transport systems perspective. Major charging modes, including conductive charging, wireless power transfer, and battery swapping, are examined in terms of their technical characteristics, deployment requirements, and suitability across different mobility contexts. The role of international standards is also considered, with emphasis on interoperability and the development of scalable, cross-regional charging networks. In addition, optimisation approaches are synthesised, focusing on charging station allocation, load management, and network coordination. These methods are discussed in relation to their capacity to improve accessibility, balance demand, and support the efficient operation of coupled transport–energy systems. Despite continued advances, several structural challenges persist, including uneven infrastructure distribution, limited standard alignment, and insufficient coordination between transport planning and energy management. Addressing these issues will be critical for enabling large-scale EV adoption and supporting the transition towards low-carbon and resilient mobility systems.

Keywords: Environment; Electric vehicles; Smart mobility, Sustainable transport systems; Charging infrastructure; Charging optimization; Infrastructure planning; Sustainable growth

1 Introduction

Over the past decade, electric vehicle (EV) technology has experienced remarkable growth, driven by its potential to address climate change, reduce dependence on fossil fuels, and improve urban air quality. The transportation sector contributes nearly 14% of global greenhouse gas (GHG) emissions, according to 2025 global emissions data, with road travel being the largest source. Among the world’s top emitters, China accounts for about 30% of total emissions, as of 2025, largely from coal-based power and heavy industry, and has pledged to peak emissions by 2030 and achieve carbon neutrality by 2060. The United States, contributing 13–15% of global emissions primarily from transport and electricity generation, has committed to cutting emissions by 50–52% by 2030 relative to 2025

levels and reaching net-zero by 2050 [1]. The European Union, responsible for around 8% of emissions, has already advanced renewable energy deployment and industrial regulation, targeting a 55% reduction by 2030 and full neutrality by 2050. India, the third-largest emitter at 7%, remains heavily reliant on coal but has committed to achieving net-zero by 2070, supported by rapid renewable energy expansion.

In this context, EVs emerge as a fundamental solution to decarbonize transport. They produce zero tailpipe emissions, significantly lowering urban pollution, while their widespread adoption by 2025 could reduce global CO₂ emissions by up to 1.5 billion metric tons annually, according to recent international energy outlook projections. Unlike internal combustion engine (ICE) vehicles, which convert only 20–30% of grid energy into motion under typical operating conditions, EVs achieve 60–70% at the drivetrain level conversion efficiency, making them far more energy-efficient. Moreover, when coupled with renewable energy, EVs strengthen energy security, reduce fossil fuel dependency, and support the global energy transition [2]. With Vehicle-to-Grid (V2G) technology, EVs can also stabilize power systems by supplying electricity back to the grid during peak demand.

The US government is aiming to generate 30% of its energy demand from solar photovoltaic systems by 2030 [3] according to recent federal clean energy transition roadmaps. However, recent developments in EV technologies between 2020 and 2025 are enhancing social benefits and boosting industrial and transportation-sector growth, whereas EV adoption remains constrained by the limitations of energy storage units (batteries), which face challenges such as capacity, size, charging/discharging rate, weight, dimensions, and cost [4]. These constraints not only increase the overall cost of EVs but also restrict their size, range, and scalability [5]. To address these challenges, many automobile manufacturers are investing heavily in next-generation battery technologies designed for longer lifespan, lower cost, faster charging, and higher energy density [6]. In spite of these efforts, recent reviews highlight the persistent challenges published between 2020 and 2025 are thermal management, degradation under fast-charging, supply chain constraints, recycling and second-life use of batteries, and grid-integration barriers associated with large-scale EV deployment. These emerging critiques underline that while the ambition to decarbonise transport and energy systems is strong, achieving it will require systemic advances in battery design, material sourcing, manufacturing scale-up, and integrated charging infrastructure [7].

It is estimated that most modern automobiles are still powered by ICE, as of 2023 global vehicle fleet statistics, which emit large amounts of carbon dioxide (CO₂) and significantly accelerate climate change. EVs running on electric motors instead of ICEs eliminate tailpipe emissions and align strongly with environmental protection goals. Alongside this shift, many energy sectors are transitioning toward renewable power sources such as solar PVs and wind energy. For example, in 2024, the average cost of lithium-ion battery packs dropped by about 20% to roughly USD 115 per kWh [7], driven by overcapacity, cheaper materials, and wider adoption of LFP battery chemistry. Despite these advances, battery storage units remain the primary limitation in EVs. Challenges include limited capacity, slower charging/discharging rates, heavy weight, high cost, and size constraints, all of which increase the overall vehicle cost and limit both design options and mass adoption [8]. For example, in India, EV sales rose to around 1.43 million in 2024, up from 1.03 million the previous year [8], helped by subsidies and incentive programs. Projections suggest that by 2026, battery pack costs could decline further toward USD 80 per kWh according to recent industry forecasts (2023–2025 reports), which could bring EVs closer to cost-parity with ICE vehicles in many markets.

The race to improve battery technology is gaining momentum worldwide, driven by the urgent need for cleaner and more efficient transportation [9]. The US government now projects an expansion of charging infrastructure to support its electric-vehicle transition. As of the end of 2024, the US had deployed about 204,000 [10] non-home public charging points, a 35% increase over 2023, but falls short of the scale required for future EV growth. Meanwhile, Australia's EV market is surging; in 2023, EV sales reached 98,409 units [11] according to 2024 national transport statistics, representing approximately 12% of all new light-vehicle sales in the country and more than doubling from the previous year. New Zealand is also taking bold steps toward sustainability [12]. By 2035, the country aims to significantly reduce its dependence on fossil fuels, replacing them with renewable energy sources (RES). Governments around the world are encouraging EV adoption through tax incentives, grants, and benefits like access to high-occupancy vehicle lanes. These measures are helping to attract a broader range of consumers and accelerate market growth [13]. The global EV powertrain market is projected to grow from 20 billion in 2025 to nearly 160 billion by 2035 [13], driven by modular designs, AI-based energy management, and wireless charging.

This study examines the latest EV architectures, evaluates their advantages and limitations, and explains how they integrate with emerging renewable energy systems such as solar, wind, hybrid microgrids, and V2G/V2X networks [14]. The study categorises modern EV charging approaches, including fast conductive charging (CC), wireless inductive charging, dynamic in-motion charging, and battery-swap systems, and assesses their technical maturity and deployment challenges. In addition, the article reviews state-of-the-art optimisation strategies designed to enhance charging efficiency, reduce operational costs, alleviate grid stress, and improve overall system reliability. These include AI-driven scheduling, load balancing, smart charging algorithms, and predictive energy-management models. The structural outline of this research article is presented in Figure 1.

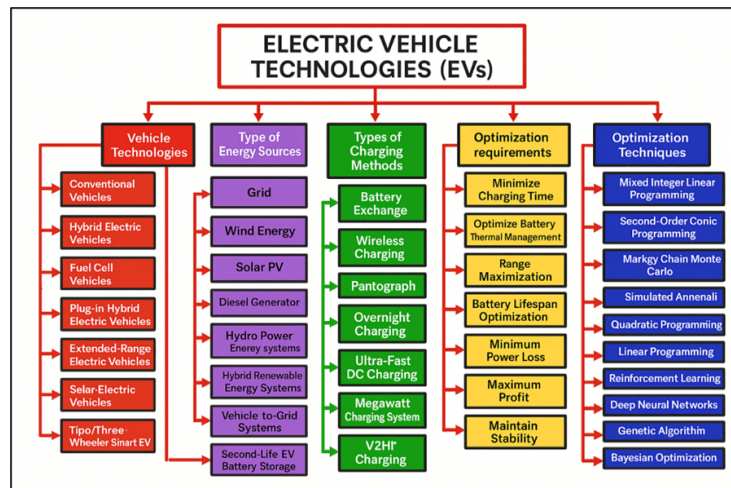


Figure 1. Recent developments in electric vehicle (EV) technologies

Source: adapted from study [15]

While several articles have examined individual aspects of EV technologies, such as battery systems, charging infrastructure, or optimization algorithms, most studies focus on a single technical domain. In contrast, this review provides an integrated perspective that connects vehicle architectures, battery management systems (BMSs), charging standards, grid interaction challenges, and multi-objective optimization frameworks within a unified sustainability context. By combining technical modelling insights with infrastructure and policy considerations, this work offers a cross-disciplinary synthesis that supports both engineering design and large-scale EV-grid deployment strategies.

2 Methodology

EVs have emerged as a central element of global strategies aimed at reducing GHG emissions, improving energy efficiency, and accelerating the shift toward sustainable mobility. Rapid progress in battery technologies, charging systems, power electronics, and energy-management architectures has transformed EV performance and feasibility in recent years. To ensure comprehensive coverage, the study draws on high-quality scientific sources, including IEEE Xplore, DOAJ, ScienceDirect, Springer, Taylor & Francis, and Google Scholar, along with government reports, industry standards, and international guidelines that reflect the latest technical advancements. The search strategy centered on key terms such as EV technologies, BMSs, fast and wireless charging, V2G integration, optimization algorithms, EV standards, battery swapping, and renewable energy integration. Only English-language publications from 2018 to 2025 were included, with priority given to peer-reviewed articles, conference papers, technical reports, and standard documents that provided substantial technical insights. The selected literature was then grouped into major thematic areas, including vehicle technologies, i.e., EV, hybrid electric vehicle (HEV), plug-in hybrid electric vehicles (PHEV), fuel cell electric vehicle (FCEV), battery systems and BMS, charging methods and standards, optimization and energy-management approaches, and comparative evaluations. Each study was assessed for its technical value, performance measures, limitations, and contribution to sustainable transportation, with special attention to charging efficiency, battery behavior, grid impact, and optimization results. By synthesizing findings across studies, the review identifies key technological trends, persistent challenges, and emerging opportunities, while also highlighting unresolved issues in battery development, wireless charging technologies, V2G integration, and multi-objective optimization that warrant further research.

To enhance transparency, the literature selection was carried out in multiple stages. Keyword combinations such as EV and “charging optimization, EV and grid integration, and V2G and energy management were applied across the selected databases for publications between 2018 and 2025. After removing duplicates and screening titles and abstracts for relevance, only peer-reviewed studies directly related to EV charging systems, grid interaction, and optimization techniques were retained. Studies that were outdated, purely policy-focused, or lacking technical and methodological clarity were excluded from the final review.

3 Vehicle Technologies

For over two centuries, fossil fuels have powered most transportation systems. While effective, this reliance has led to high CO₂ emissions and rising fuel costs. EVs offer a promising alternative by cutting both emissions and fuel expenses. As fossil fuel reserves continue to decline, the need for cleaner energy sources becomes more urgent, driving innovation in automotive technology. Today’s vehicles fall into four main categories: conventional, hybrid electric, fuel cell, and fully electric. Conventional vehicles run solely on fossil fuels and are major contributors to

carbon pollution. HEVs, on the other hand, combine a combustion engine with a battery system. This dual-source setup allows them to operate using either or both power sources, resulting in lower CO₂ emissions compared to traditional vehicles.

3.1 Clean Energy Vehicles: Fuel Cells, Battery Electric Vehicles, and Hybrids

Fuel cell vehicles (FCVs), battery electric vehicles (BEVs), and HEVs represent different pathways toward low-emission transportation. FCVs generate electricity onboard using hydrogen fuel cells, emitting only water vapour [16]. However, their large-scale deployment is constrained by high hydrogen production costs, limited refuelling infrastructure, and storage challenges, restricting practical implementation to regions with established hydrogen networks. BEVs have experienced rapid technological progress in range, charging speed, and energy density, supporting their widespread adoption [17]. Nevertheless, large-scale BEV deployment depends heavily on the availability of fast-charging infrastructure, grid capacity, and critical mineral supply chains. Long-distance travel requires high-power corridor charging networks, making infrastructure planning a key enabling factor.

HEVs combine ICEs with electric motors to improve fuel efficiency, particularly in urban stop-and-go conditions. While they reduce fuel consumption, their continued reliance on fossil fuels limits their long-term decarbonization potential compared to fully electrified systems. Figure 2 illustrates the major vehicle technology pathways [18]. From an infrastructure perspective, BEVs currently impose the most significant demand on charging network expansion and grid coordination, thereby motivating the need for integrated transportation–energy planning frameworks.

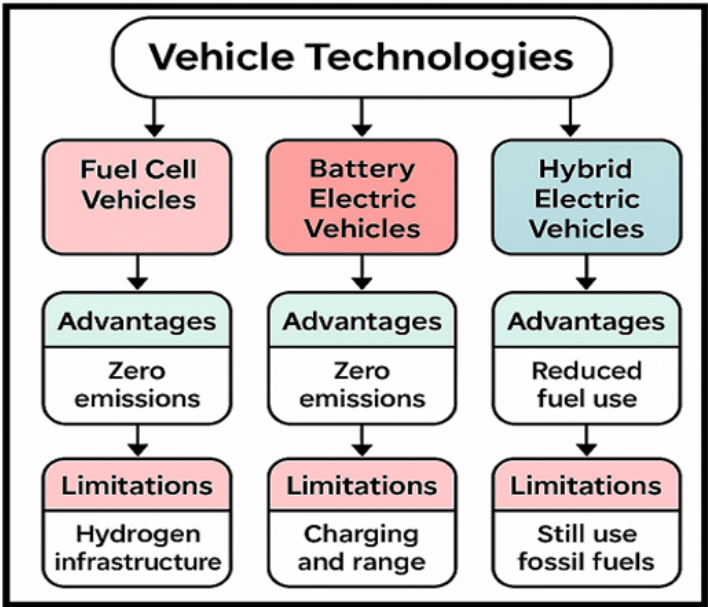


Figure 2. Vehicle technologies

3.2 Critical Analysis on Vehicle Technologies

In regions where charging infrastructure remains limited, HEVs provide a practical transitional solution. By combining an ICE with an electric motor, HEVs ensure operational continuity while improving fuel efficiency through regenerative braking and adaptive power management. Compared to conventional vehicles, HEVs reduce fuel consumption and GHG emissions, particularly in congested urban traffic. Modern HEV architectures, including series, parallel, and series-parallel configurations, incorporate higher-voltage systems, advanced power electronics, and intelligent control strategies to enhance efficiency. Recent developments, such as silicon carbide inverters and predictive torque management, have improved urban fuel economy by up to 15% and reduced CO₂ emissions by approximately 25% compared to early-generation systems. However, each architecture involves trade-offs: series hybrids offer strong urban efficiency but multiple energy conversion stages; parallel hybrids perform better at highway speeds but require complex torque coordination; and series-parallel systems balance performance at the expense of higher control complexity. These design considerations continue to influence optimization strategies and infrastructure compatibility.

Figure 3 illustrates the energy flow characteristics of hybrid configurations. From an infrastructure and system-level perspective, BEVs and FCEVs impose distinct charging and refuelling demands, reinforcing the need for integrated transportation–energy planning frameworks.

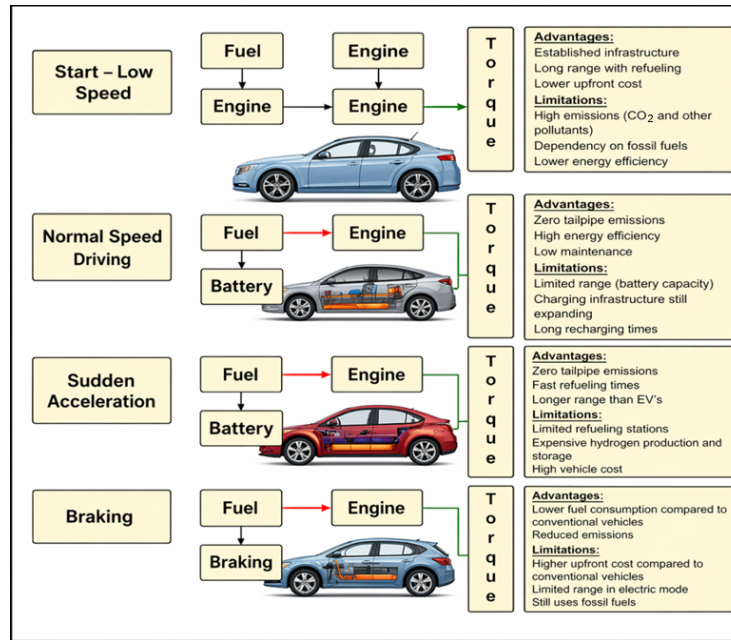


Figure 3. Power flow and torque contribution of hybrid electric vehicles (HEVs) during start, normal driving, acceleration, and braking

Note: EV (electric vehicle).

4 Types of Batteries for Electric Vehicles

EVs depend entirely on battery packs to store and deliver energy to their motors. Commercially available chemistries vary in energy density, lifespan, cost, safety, and power output [19, 20]. Choosing the right battery involves balancing range, durability, and price. Table 1 summarizes the key characteristics of today's leading EV battery types and emerging options.

Table 1. Key characteristics of today's leading EV battery types and emerging options

Battery Type	Energy Density (Wh/kg)	Cycle Life	Cost	Key Advantages	Limitations	Typical Applications
Lead-Acid	30–50	200–500	Low	Mature, low-cost, recyclable	Heavy, low energy density	Entry EVs, backup systems
NiMH	60–120	500–1,000	Medium	Durable, moderate cost	Self-discharge, moderate density	Hybrid vehicles
LFP	90–160	2,000–5,000	Medium–low	High thermal stability, long life	Lower energy density	Mass-market EVs, buses
NMC	150–220	1,000–2,000	High	High energy density, flexible design	Shorter lifespan vs LFP	Mid/high range EVs
NCA	200–260	1,000–2,000	High	Very high density, fast charging	Higher cost, thermal sensitivity	Performance EVs
Solid-State	300–500	1,000–3,000	Very high	Superior safety, high density	Manufacturing challenges	Prototype/next-generation EVs

Note: EV (electric vehicle); NiMh (Nickel metal hydride); LFP (Lithium-iron-phosphate); NMC (Nickel manganese cobalt); NCA (Nickel cobalt aluminium)

4.1 Intelligent Battery Management Systems

The BMS monitors cell voltage, temperature, and charge levels to prevent overcharging, deep discharge, and thermal instability [21]. It incorporates thermal regulation and cell balancing mechanisms, either passive or active, to

maintain uniform cell performance and extend battery lifespan. These functions ensure operational safety and stable power delivery under varying load conditions. In addition to protection, the BMS estimates State of Charge (SoC) and State of Health (SoH) to support range prediction and energy management. From an infrastructure perspective, accurate SoC estimation and adaptive charge–discharge control directly influence charging demand profiles, peak load formation, and grid interaction strategies [22, 23]. Advanced control approaches have demonstrated measurable efficiency improvements under intelligent charging coordination scenarios. Figure 4, illustrates the performance characteristics of advanced battery technologies in terms of energy density and specific energy.

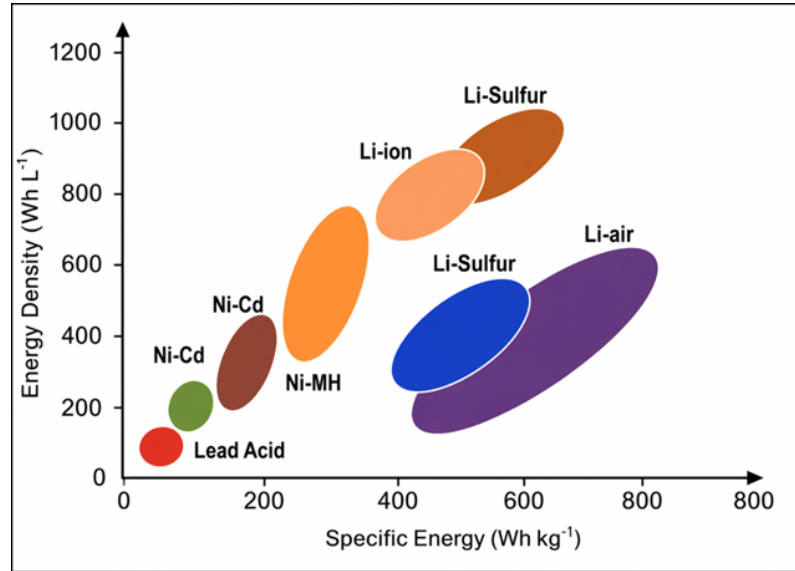


Figure 4. Characteristics of the advanced battery technologies [6]

4.2 Charging Methods

EVs today can be recharged through three main approaches: battery swapping, CC, and wireless charging [24]. In the battery swapping method, a depleted pack is physically exchanged for a fully charged one at a swapping station in just a few minutes, while the removed pack is recharged slowly behind the scenes. CC relies on a direct wired connection and splits into two variants: pantograph charging, which uses an overhead arm to connect rapidly, often favoured for buses and large fleet vehicles, and overnight depot charging, where vehicles plug in for longer, lower-power top-ups [25, 26]. Wireless charging eliminates cables altogether by using electromagnetic induction pads embedded in floors or roadways, letting any compatible vehicle recharge simply by parking over the pad [27]. Each technique carries its own trade-offs. Battery swapping delivers near-instant “fill-ups” and can integrate renewable-powered charging stations, but it demands standardized packs, significant station investment, and ample spare batteries to meet peak demand [28, 29]. Conductive pantograph systems offer ultra-fast power delivery yet require specialized infrastructure and raise safety and maintenance concerns, whereas depot charging is inexpensive to install but ties vehicles down for hours.

Wireless charging for EVs relies on electromagnetic induction, which is a primary coil buried in the roadway or embedded pad that generates an alternating magnetic field that induces current in a secondary coil mounted beneath the vehicle, so drivers can recharge simply by parking or even driving over the pad without handling cables [30, 31]. Since no physical connection is needed, this approach works across different battery standards and vehicle models, with charging power determined by the secondary coil’s design. Efficiency is highest when the coil gap stays between 20 cm and 100 cm, but it can drop due to eddy currents and misalignment.

Wireless power transfer (WPT) methods fall into three distance-based categories they are near-field WPT (millimeter-to-centimeter range) [32], which offers the best efficiency and includes capacitive coupling, inductive coupling, and resonant inductive coupling (tuning transmitter and receiver to the same frequency for stronger, more forgiving transfers ideal for dynamic charging); medium-field WPT, which employs a series of aligned intermediate coils to span larger distances; and far-field WPT, which beams energy via microwaves or laser light to PV receivers for long-range, remote, or space applications despite its lower transfer efficiency [33]. Continuous innovations, such as dynamic transmitter coils that adapt to differing receiver frequencies, anti-misalignment compensation, and integrated communication protocols to prevent interference, are overcoming losses and paving the way for practical, scalable wireless and in-motion EV charging networks. Figure 5 illustrates the global growth of public EV charging infrastructure between 2015 and 2024.

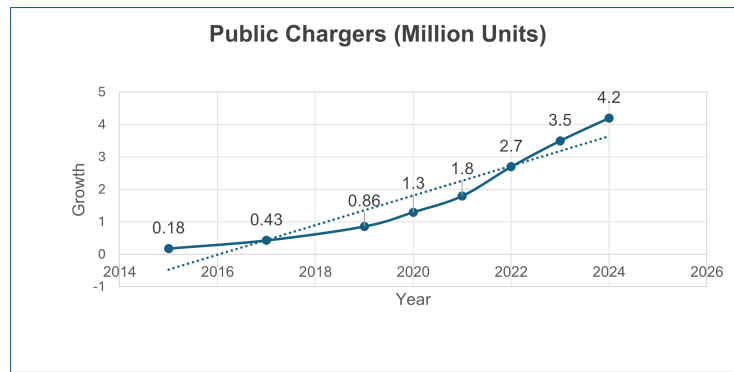


Figure 5. Global public charging infrastructure growth [34]

In CC, the vehicle simply plugs into the power grid through a dedicated inlet and power-conditioning unit. Charging setups are divided into three standardized levels, L1, L2, and L3, each defined by different voltage [35], current, and power ratings to suit various vehicle and battery specifications. Unlike lower-level charging, L3 fast chargers deliver much higher power but can introduce voltage deviations and reactive power swings in the distribution network, leading to increased losses and potential stress on both batteries and charging hardware. Conductive systems also enable bi-directional power flow, supporting V2G applications that require robust communication protocols but may accelerate battery ageing if cycled frequently [36]. For heavy-duty fleets such as buses, two practical conductive solutions have emerged: overnight depot charging, which replenishes large battery packs slowly during off-peak hours to extend battery life, and pantograph charging, which uses either top-down or bottom-up robotic arms to deliver rapid, high-power fills with lower upfront battery costs but higher infrastructure investment. Figure 6 presents the historical decline in global lithium-ion battery pack prices (2010–2024). The long-term downward trend highlights improved manufacturing scale, technology advancements, and cost competitiveness of EV systems.

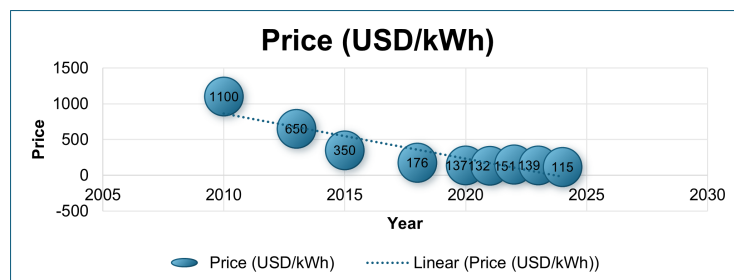


Figure 6. Battery cost reduction trend [37]

The battery exchange method (BEM) offers fast turnaround by swapping depleted packs for charged ones in just minutes, making it ideal for drivers seeking minimal downtime and for fleets that need rapid refueling [38]. Its ability to leverage V2G and Grid-to-Vehicle (G2V) flows helps balance power demand and supply, and stations can link directly with RES, cutting overall carbon footprints. Swapping also reduces stress on cells by enabling slow, controlled charging off-site, which can extend battery life. However, BEM requires large inventories of standardized batteries and substantial station footprints, driving up capital and operational costs. Keeping enough charged packs on hand to meet peak demand is challenging, and the high upfront investment in batteries and swapping infrastructure can deter widespread adoption.

4.3 Vehicle-to-Grid Challenges

Implementing V2G services accelerates battery wear by introducing extra charge-discharge cycles that can shorten pack life and undermine owner confidence, driving ongoing research into advanced BMS algorithms to mitigate degradation. True bidirectional power flow also depends on specialized V2G chargers, which currently cost two to three times more than unidirectional units and remain scarce in the public charging landscape [39]. Moreover, the inherently unpredictable availability of vehicles driven by diverse driving patterns and charging behaviors complicates fleet-level aggregation and makes it difficult to guarantee reliable grid support or revenue streams for EV owners.

Effective V2G deployment requires a smarter grid infrastructure capable of handling dynamic reversals of power flow, managing voltage fluctuations, harmonics, and reactive power in real time, upgrades that many utilities have yet to plan or fund. While communication standards such as ISO 15118 lay the groundwork for secure, seamless

EV-grid interactions [40], the lack of fully harmonized protocols and cybersecurity safeguards raises concerns over interoperability and data integrity. Regulatory and business frameworks remain fragmented across regions, lacking transparent compensation models or clear rules for energy trading, which stifles investment and slows the scaling of V2G ecosystems [41].

While the preceding sections have examined vehicle technologies, battery systems, charging mechanisms, and grid-interaction challenges, the increasing scale of EV deployment introduces significant operational complexity [42]. As EVs become deeply integrated into transportation infrastructure and power distribution networks, technical advancements alone are insufficient to guarantee efficiency, stability, and economic feasibility [43]. The simultaneous interaction between charging demand, grid capacity constraints, renewable energy variability, and user behavior necessitates systematic decision-making frameworks. Therefore, optimization emerges as a natural extension of the technical discussion.

4.4 Transportation-Oriented Charging Infrastructure Planning

Recent transportation-oriented research demonstrates that EV charging infrastructure planning must be grounded in traffic flow dynamics and realistic user behavior modeling rather than treated as an isolated energy allocation problem. For instance, a comprehensive review of EV charging demand and distribution models incorporating driver behavior, identifying that stochastic route choice, heterogeneous SoC thresholds, and non-uniform charging preferences significantly affect spatial charging demand patterns [44]. Their analysis shows that models ignoring behavioral variability risk mis-estimating station utilization and queue formation, particularly in high-penetration scenarios. Within urban environments, a systematic literature review [45] of 57 studies on charging infrastructure planning concluded that most existing models rely on static or flow-based formulations, with limited incorporation of multi-period demand evolution and geometric segmentation. Their findings emphasize the necessity of agent-based and temporally adaptive planning frameworks to capture traffic density shifts and prevent partial spatial coverage of charging networks.

For motorway and transit corridors [46], a hybrid modeling framework combining probabilistic traffic-flow-based demand estimation with discrete-event simulation to analyze fast-charging hub performance. Using seasonal traffic intensity data, projected EV adoption rates, and battery characteristics, their study demonstrates that peak-hour charging demand and queue dynamics are directly correlated with directional traffic volume. This highlights that charging hub sizing and grid connection capacity must be aligned with transport corridor characteristics rather than solely average load assumptions.

Moreover, recent work on coupled traffic power systems further strengthens this perspective. A review on integrated power distribution and transportation network models [47], showing that charging scheduling, navigation, and station planning, are interdependent processes influenced by congestion, electricity pricing, and user heterogeneity. Similarly, some studies use stochastic user equilibrium theory in power-transportation coupled optimization, illustrating that EV route and charging decisions follow probabilistic equilibrium behavior rather than fully rational deterministic selection. Their framework demonstrates that incorporating stochastic traffic equilibrium improves both congestion mitigation and distribution network stability.

Collectively, these studies establish that infrastructure-level optimization must integrate traffic forecasting, stochastic user equilibrium principles, and queue dynamics to ensure system resilience and balanced utilization. Therefore, EV charging coordination should be conceptualized as a transportation-integrated infrastructure planning problem embedded within intelligent transportation systems, rather than solely as a grid-constrained energy dispatch model.

5 Electric Vehicle Charge Scheduling Process

5.1 Transportation-System-Level Simulation Perspective

Optimization is a key approach that seeks to make systems, architectures, or decisions as efficient as possible within defined limitations, usually by minimizing costs or maximizing benefits. In the context of EVs, optimization plays a central role in managing charging infrastructure, energy use, battery life, and integration with the electrical grid. The scheduling of EV charging involves balancing multiple factors such as travel routes, charging duration, dynamic electricity prices, and the environmental footprint of using RES. Recent studies highlight that advanced optimization methods [48], such as artificial intelligence (AI) and machine learning (ML)-based algorithms, are increasingly applied to schedule charging in real time, reducing peak load stress on grids and enhancing renewable energy utilization. For example, V2G integration combined with smart charging can flatten demand curves and provide grid stability services, while minimizing battery degradation. Bagdadee et al. [49] examined design and technical constraints, highlighting the potential for solar power to improve urban transportation infrastructure with cleaner and more resilient alternatives.

The transportation sector is widely recognized as a major contributor to global GHG emissions, making emission reduction strategies an important research priority. EVs are frequently presented as a promising alternative to

conventional ICE vehicles due to their zero tailpipe emissions and potential to improve urban air quality. However, the overall environmental benefit of EVs depends strongly on the electricity generation mix and lifecycle emissions associated with battery production. While EVs can operate on RES such as solar and wind, the extent of these benefits varies across regions depending on renewable penetration and grid carbon intensity. From an energy efficiency perspective, electric motors typically achieve higher conversion efficiency than ICE engines, although the overall system efficiency also depends on charging infrastructure, power electronics, and battery management strategies. Technological advancements and simplified drivetrain architectures have contributed to lower maintenance requirements and improved reliability compared to conventional vehicles. Market trends further illustrate the rapid growth of EV adoption; global EV sales increased by nearly 50% between 2022 and 2025 [50], with projections suggesting that EVs could exceed 20% of new car sales worldwide by 2025 [51]. This expansion is supported by declining battery costs, policy incentives, and infrastructure investments, although the long-term sustainability of these policies and supply chain limitations for critical battery materials remain ongoing concerns.

Optimization of EV systems has been widely investigated as a means of improving operational efficiency, reducing charging costs, and enhancing user experience. Various strategies, including advanced charging scheduling, intelligent routing, and improved battery management systems, have been proposed to reduce energy consumption and extend battery lifespan. Smart charging combined with Time-of-Use (ToU) pricing has demonstrated potential for shifting charging demand to off-peak periods, thereby lowering electricity costs and reducing grid stress. Recent pilot projects conducted between 2023 and 2025 indicate that coordinated charging strategies can reduce consumer electricity expenses by approximately 25–30% under favourable tariff structures [52]. Nevertheless, the effectiveness of these approaches depends on several practical conditions, including grid capacity, communication infrastructure, and user participation rates. V2G integration has also been explored as a mechanism for improving grid stability by enabling EVs to supply power during peak demand periods and support renewable energy balancing. However, widespread deployment of V2G remains constrained by regulatory uncertainty, battery degradation concerns, and the need for bidirectional charging infrastructure.

Consequently, while optimization frameworks demonstrate promising results in controlled environments and pilot studies, large-scale implementation requires further evaluation under realistic operational conditions. In spite of these potential advantages, several technical, financial, and infrastructural challenges limit the effectiveness of EV system optimization. One major constraint is the intermittent nature of RES, such as solar and wind, whose variability complicates the alignment between renewable generation and charging demand [53]. Effective optimization also requires robust infrastructure, including advanced charging networks, smart grids, real-time data analytics, and reliable communication systems. In regions with underdeveloped grid infrastructure, the deployment of such technologies can involve substantial investment and regulatory coordination. The integration of EVs with smart grids introduces additional operational complexity, as coordination is required between vehicles, charging stations, and distribution network operators. High installation costs associated with fast-charging stations, control systems, and grid reinforcement may further slow infrastructure expansion despite long-term economic benefits.

A wide range of computational techniques has been applied to EV optimization problems, particularly for multi-objective scenarios involving charging scheduling, energy management, and grid interaction. Nature-inspired algorithms such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO) have been widely used due to their flexibility in handling complex, nonlinear search spaces. GA approaches emulate evolutionary processes to iteratively improve candidate solutions, making them suitable for scheduling and routing problems. PSO algorithms model collective behavior in swarm systems and have been applied to large search-space optimization tasks such as coordinated charging management. Similarly, Ant Colony Optimization (ACO) algorithms use pheromone-based learning mechanisms to identify efficient routing or scheduling paths [54, 55]. Other advanced techniques, including Chaotic Harris Hawk Optimization (CHHO), attempt to improve global exploration capabilities and reduce the risk of premature convergence. Deterministic optimization methods such as Mixed Integer Linear Programming (MILP) offer mathematically rigorous formulations and can effectively incorporate grid constraints and ToU pricing structures [56]. More recently, machine learning-based methods, particularly deep reinforcement learning (DRL), have been explored for adaptive charging strategies under uncertain and dynamic conditions [57]. However, each class of methods presents trade-offs in terms of computational complexity, scalability, and real-time feasibility. Consequently, selecting an appropriate optimization strategy requires careful consideration of system scale, data availability, and infrastructure constraints rather than relying on a single universally optimal approach.

5.2 Maximize Grid Stability

Maximizing grid stability involves ensuring that the supply from the grid matches the total electrical demand, including EV charging loads and other household or industrial loads. With the increasing penetration of RES such as solar and wind, which naturally fluctuate, maintaining stability becomes essential. Large variations in RES output can lead to frequency deviations, voltage fluctuations, and even grid failure if not properly balanced [58].

The objective of the grid-stability function is therefore to maintain the balance between grid power supply and EV charging demand at every time instance.

Objective Function: Maximize Grid Stability

$$\text{Maximize} = \sum_{t=1}^N [P_{grid}(t) - P_{EV}(t)] \quad (1)$$

where, $P_{grid}(t)$ = Power supplied by the grid at time t , $P_{EV}(t)$ = Total EV charging power demand at time t , and N = Total number of time intervals.

In practical power systems, deviations between supply and demand can lead to frequency fluctuations and voltage instability. The formulation, therefore, represents a load-balancing objective aimed at minimizing power mismatch over time, which supports stable grid operation under increasing EV penetration [59].

5.3 Constraints

5.3.1 Power balance constraint

The power supplied by the grid should always meet or exceed the EV charging power demand to maintain balance.

$$P_{grid}(t) \geq P_{EV}(t) \quad (2)$$

This prevents shortages and ensures continuous, reliable charging.

5.3.2 Maximum grid capacity constraint

The grid cannot be overloaded beyond its maximum operating capacity. This constraint prevents grid collapse.

$$P_{grid}(t) \leq P_{grid}^{\max} \quad (3)$$

where, $P_{grid}^{\max}$ is the highest load the grid can safely supply.

5.3.3 Time-of-Use pricing constraint

Electricity pricing varies throughout the day based on demand. ToU pricing encourages users to charge EVs during off-peak hours (night or midday solar peak), reducing grid stress.

$$C_{charge}(t) = C_{base} + C_{peak}(t) \quad (4)$$

where, $C_{charge}(t)$ = Cost of charging at time t , C_{base} = Base tariff, and $C_{peak}(t)$ = Additional cost during peak periods.

5.4 Transmission Power Loss Modeling

Transmission and distribution lines suffer resistive losses (I^2R losses). Minimizing these losses ensures more delivered power reaches EVs effectively.

Objective Function: Minimize Power Loss

$$\text{Minimize } L = \sum_{t=1}^N \frac{I(t)^2 R}{V_{transmission}} \quad (5)$$

where, $I(t)$ = Current flowing at time t , R = Line resistance, and $V_{transmission}$ = Transmission voltage.

5.5 Power Loss Constraints and Voltage Stability

As electrical power flows through transmission and distribution networks, a portion of it is lost due to the inherent resistance of the conductors [60]. These resistive losses, often referred to as I^2R losses, increase the temperature of the lines and reduce the amount of power that ultimately reaches EV charging stations. Since losses rise with the square of the current, simultaneous charging of multiple EVs significantly increases current flow, resulting in higher line losses. In addition, the voltage level of the transmission system directly affects power losses. High-voltage systems transmit power over long distances more efficiently, as they allow the same power to be delivered with lower current, thereby reducing I^2R losses. As public EV charging infrastructure continues to expand, maintaining appropriate transmission voltage becomes increasingly important to keep the system efficient and stable.

Objective Function: Minimize Power Loss

$$\text{Minimize } L = \sum_{t=1}^N \frac{I(t)^2 R}{V_{trans}} \quad (6)$$

5.5.1 Current limitation

To prevent overheating and equipment damage, the current flowing through transmission lines must remain within their safe operating limits.

$$I(t) \leq I_{\max} \quad (7)$$

5.5.2 Voltage stability

Stable voltage ensures reliable power delivery and prevents charging interruptions. The transmission voltage must not fall below a minimum threshold.

$$V_{trans}(t) \geq V^{\min} \quad (8)$$

5.5.3 Power delivered to electric vehicle (EVs)

The actual power available to EVs is the grid-supplied power minus the system losses.

$$P_{delivered}(t) = P_{grid}(t) - L(t) \quad (9)$$

Minimizing losses increases the usable power at charging stations and improves overall system performance [61].

5.6 Minimize Waiting Time

Reducing waiting time at EV charging stations enhances user satisfaction and is essential for widespread EV adoption. Long delays diminish convenience, particularly in urban areas with high vehicle density, and can cause congestion at charging facilities.

The objective is to minimize the difference between the EV's arrival time and the time at which charging begins.

Objective Function: Minimize Waiting Time

$$\text{Minimize } W = \sum_{i=1}^{N_{EV}} (t_i^{\text{start}} - t_i^{\text{arrival}}) \quad (10)$$

5.6.1 Charging station capacity

The number of EVs charging at any moment must not exceed the physical capacity of the station.

$$N_{charging}(t) \leq N^{\max} \quad (11)$$

5.6.2 Minimum state of charge priority

Vehicles with critically low battery levels require priority to prevent complete depletion [62].

$$\text{SoC}_i(t) \leq \text{SoC}^{\min} \quad (12)$$

This ensures that EVs with insufficient charge gain access before vehicles with adequate remaining capacity.

5.6.3 Charging duration constraint

Charging time depends on battery size, charger power level, and allowable charging current.

$$t_{charge} = \frac{E_{battery}}{P_{charger}} \quad (13)$$

Optimizing charging duration prevents idle occupancy of chargers, thereby reducing queue buildup and overall waiting time.

5.7 Minimize Charging Cost

Minimizing charging cost is essential for both EV users and grid operators. By applying ToU pricing, the charging process can be shifted to low-demand periods, reducing overall energy costs [63]. ToU pricing varies throughout the day; rates are lower during off-peak hours and higher during high-demand periods, encouraging cost-efficient charging behavior [64].

Charging Time Formula (Corrected & Clear)

Charging time for each EV depends on its battery capacity, state of charge, and the power rating of the charging station:

$$t_i^{charge} = \frac{C_i^{battery} \times (\text{SoC}^{\max} - \text{SoC}_i(t))}{P_{station}} \quad (14)$$

where, $C_i^{battery}$ = Battery capacity of vehicle i , SoC^{max} = Maximum target state of charge, $SoC_i(t)$ = Current SoC at time t , and $P_{station}$ = Charging station power.

Objective Function: Minimize Total Charging Cost

$$\text{Minimize } C_{total} = \sum_{i=1}^{N_{EV}} \left(C_i^{charge}(t) \times t_i^{charge} \right) \quad (15)$$

where, $C_i^{charge}(t)$ = Time-dependent ToU price and t_i^{charge} = Charging duration.

5.7.1 Time-of-Use price limits

To ensure charging does not occur during extremely high-price durations:

$$C_i^{charge}(t) \leq C^{max} \text{ if } t \in \text{peak hours} \quad (16)$$

5.7.2 Charging power constraint

The EV charging power must not exceed the maximum allowable limit of the station.

$$P_{EV}(t) \leq P_{station}(t) \quad (17)$$

This prevents equipment overload and avoids unnecessary cost escalation.

5.8 Combined Multi-Objective Optimization

Since EV charging involves several conflicting objectives, grid stability, minimum power loss, reduced waiting time, and lower charging cost, a multi-objective optimization framework is required [65].

A weighted-sum formulation is commonly used:

$$\text{Minimize } F = w_1 \cdot \frac{1}{S} + w_2 \cdot L + w_3 \cdot W + w_4 \cdot C_{total} \quad (18)$$

where, w_1, w_2, w_3, w_4 are weights assigned to each objective, S = Grid stability measure, L = Power loss, W = Waiting time, and C_{total} = Total charging cost.

5.8.1 State of charge limits

Ensures battery health and prioritizes low-SoC vehicles.

$$SoC^{min} \leq SoC(t) \leq SoC^{max} \quad (19)$$

5.8.2 Overall power balance

If you meant the grid energy or total grid contribution over the period 1 to T .

$$E_{grid} = \sum_{t=1}^T (P_{EVs}(t) + L(t)) \quad (20)$$

The above equation ensures that the grid supplies enough energy to match EV demand and system losses.

The mathematical formulations presented in this section are not intended for numerical computation but serve to define the theoretical framework used for evaluating EV charging performance and grid interaction [66]. These objective functions and constraints provide a generalized representation of how grid stability, power loss, waiting time, and charging cost can be optimized in large-scale EV charging environments. By expressing these parameters mathematically, the review highlights the fundamental relationships between grid supply, charging demand, transmission losses, station capacity, and dynamic pricing. This formulation helps compare different optimization strategies adopted in recent research and illustrates how multi-objective models are constructed in smart charging studies [67, 68]. Although no numerical calculations are performed in this paper, the provided equations offer a unified structure that researchers can adapt for simulations, algorithm development, and further analytical studies. Thus, the mathematical models enhance conceptual clarity and support a deeper understanding of the operational challenges and optimization goals in EV charging systems.

5.9 Summary of Various Optimization Methods

A comprehensive review of optimization techniques for EV charging systems and energy management strategies highlights several algorithms, each with unique advantages, limitations, and use cases [69, 70]. Finite Horizon Scheduling (FHS), commonly applied in building-integrated microgrid (MG) systems, can satisfy up to 85% of energy demand via PV systems and effectively manage uncertainties, though it requires improvements in forecasting accuracy and does not support bidirectional energy flow. Stochastic Integer Programming (SIP) enhances system security by balancing robustness and economic efficiency, but its handling of stochastic parameters such as variable power availability and electricity prices can be challenging, particularly in EV-grid integration [71]. PSO, often used in residential energy systems, maintains user comfort with minimal energy consumption but requires integration of market information and economic analysis for full effectiveness. MILP efficiently manages the state-of-charge (SoC) of EVs and reduces electricity costs, although it performs best when combined with demand response programs and smart-home integration. Overall, these methods demonstrate that no single algorithm universally outperforms others, but instead, careful selection and hybridization are necessary to balance efficiency, cost, grid stability, and user requirements in EV charging optimization [69].

To support widespread adoption of EVs, robust energy policies and optimization strategies are essential. Model Predictive Control (MPC) combined with Optimal Control with Minimum Cost and Flexibility (OCCF) enhances the flexibility of EV charging stations [72], and reduces operational costs, although its effectiveness is limited by the lack of real-world EV usage data and integration with RES. GA demonstrates the benefits of V2G systems in reducing costs and improving energy efficiency, but their flexibility can be constrained by user-defined parameters. Linear programming optimizes simultaneous EV charging and maximizes self-sufficiency, making it particularly relevant for autonomous energy systems [73]. Collectively, these methods highlight that while advanced optimization techniques can enhance EV integration, their success depends on supportive policies, renewable energy integration, and ongoing research to balance technical feasibility, economic efficiency, and grid reliability.

Advanced optimization techniques play a critical role in maximizing renewable energy integration, improving grid stability, and reducing costs in EV systems. MILP efficiently maximizes the use of renewable energy sources (RES), minimizes imported energy, and can inject surplus power into the grid, though its application to larger systems requires performance comparisons with other methods. NSGA-II effectively reduces GHG emissions and achieves energy savings [68], although its high computational requirements can limit scalability for multi-objective problems. Mixed Integer Nonlinear Programming (MINLP) lowers energy costs and GHG emissions based on ToU tariffs, but it does not fully address uncertainties in EV driving patterns and electricity pricing. The Artificial Bee Colony (ABC) algorithm enhances system efficiency and security while lowering total costs, though additional validation is required for complex hybrid energy storage scenarios. Collectively, these methods illustrate that no single approach can meet all EV integration, energy management, and grid stability requirements; each technique offers distinct advantages and limitations, highlighting the importance of hybrid and adaptive strategies to optimize performance across diverse operational contexts. Figure 7 provides a comparative overview of reported performance improvements achieved by major EV charging optimization algorithms. The figure highlights differences in cost reduction, peak-load mitigation, and system efficiency enhancement.

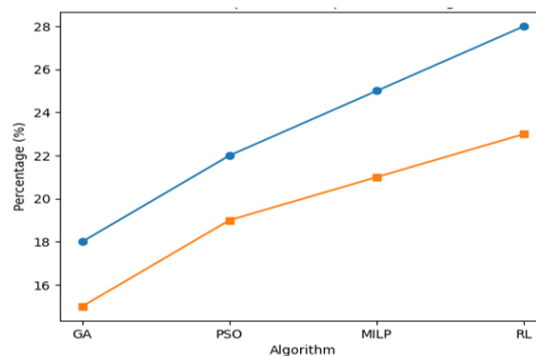


Figure 7. Optimization algorithm performance comparison [74]

Note: GA (Genetic Algorithms), PSO (Particle Swarm Optimization), MLP (Multilayer Perceptron), RL (Reinforcement Learning).

From a practical implementation perspective, deterministic methods such as MILP provide precise solutions but may face scalability limitations as system size increases. Metaheuristic approaches such as PSO and GA improve flexibility [75], but can require parameter tuning and higher computational time. Deep reinforcement learning (DRL) methods offer adaptive real-time capability, yet depend heavily on training data and computational resources. Therefore, recent research increasingly explores hybrid approaches that balance solution accuracy with real-time

feasibility.

5.10 Illustrative Case Study

To demonstrate the practical effectiveness of the proposed multi-objective framework, a simplified transportation corridor scenario is considered.

- Three EV charging stations (S1, S2, S3);
- Total of 50 EVs arriving during a 2-hour peak period;
- Each EV requires 18 kWh on average;
- Maximum feeder capacity limit: 500 kW;
- Charging power per vehicle: 22 kW (fast AC charging).

Objectives:

- Minimize power loss;
- Minimize peak load deviation; and
- Minimize average waiting time.

Two scenarios are compared:

- Uncoordinated charging (first-come-first-served);
- Coordinated charging using the proposed optimization framework.

Under uncoordinated charging (from Table 2), simultaneous vehicle demand exceeds feeder capacity by approximately 18%, resulting in voltage deviation and increased distribution losses [76]. Queue formation increases waiting time and reduces service efficiency.

Table 2. Performance metric comparison between uncoordinated and optimized electric vehicle (EV) charging scenarios

Performance Metric	Uncoordinated	Optimized	Improvement
Peak load (kW)	590 kW	480 kW	↓ 18.6%
Average waiting time	24 min	14 min	↓ 41.7%
Power loss (kW)	42 kW	31 kW	↓ 26.2%
Feeder overload occurrence	Yes	No	Eliminated

Note: NiMh (Nickel metal hydride); LFP (Lithium-iron-phosphate); NMC (Nickel manganese cobalt); NCA (Nickel cobalt aluminium).

When the proposed multi-objective optimization is applied, charging schedules are staggered to remain within feeder capacity limits, resulting in a balanced load distribution across stations. This allocation reduces the queue lengths and waiting times, which leads to lower distribution power losses, which improves the system stability.

This simplified case demonstrates that integrating traffic arrival patterns with grid constraints can significantly enhance both transportation service quality and electrical network stability. The results confirm the practical applicability of the proposed framework within intelligent transportation infrastructure contexts.

5.11 Systematic Evaluation and Comparative Analysis

To avoid a purely descriptive presentation, the reviewed EV charging and optimization strategies can be systematically evaluated based on four core criteria: (i) computational complexity, (ii) scalability under high EV penetration, (iii) real-time feasibility, and (iv) infrastructure adaptability.

Rule-based strategies exhibit low computational burden and high real-time feasibility but lack adaptability under dynamic traffic and grid conditions. Metaheuristic algorithms (e.g., GA, PSO) demonstrate strong global search capability and flexibility but may suffer from increased computational time when applied to large-scale transportation networks [77]. Convex and mixed-integer programming approaches offer mathematical rigor and deterministic convergence, yet scalability becomes challenging under stochastic traffic-flow coupling. Learning-based methods, including reinforcement learning, provide adaptive performance in uncertain environments but require significant training data and may introduce stability concerns during deployment.

From an infrastructure perspective, approaches integrating traffic flow dynamics and feeder constraints outperform purely grid-centric optimization models in reducing congestion and peak loading. However, practical implementation requires balancing algorithm sophistication with computational efficiency and regulatory constraints. This structured evaluation clarifies that no single optimization strategy is universally superior; rather, suitability depends on system scale, data availability, and infrastructure maturity [78].

6 Future Research Directions

Future research should prioritize the development of traffic-aware charging infrastructure models that integrate vehicle arrival rates, route selection behavior, and peak-hour variability into station placement and sizing decisions.

Corridor-level fast-charging planning, particularly along highways and high-density urban networks, requires stochastic demand forecasting and queue modeling to prevent congestion and feeder overloading. Coupled transportation–power network frameworks should be further refined to support coordinated infrastructure expansion under high EV penetration. As EV adoption increases, charging coordination must evolve toward scalable, real-time optimization algorithms capable of handling large fleets under dynamic grid constraints. Multi-objective formulations should incorporate feeder capacity limits, voltage stability, electricity pricing, and user waiting time simultaneously. Future research should focus on distributed and decentralized optimization methods that reduce computational burden while maintaining system stability in high-penetration scenarios.

Bidirectional charging and V2G systems require further investigation to quantify their economic viability, battery degradation impact, and contribution to voltage regulation and peak shaving. Dynamic and wireless charging technologies also demand system-level evaluation in terms of infrastructure cost, efficiency, and electromagnetic compatibility. Future studies should assess how these technologies integrate with intelligent transportation systems and smart-grid platforms.

While battery performance improvements remain important, future research should focus primarily on parameters influencing infrastructure demand, including charging rate capability, degradation under fast charging, and second-life applications for grid storage. Integrated battery-health-aware charging strategies can enhance both grid coordination and asset longevity. Life-cycle assessment methodologies should also evaluate environmental impacts under large-scale deployment conditions. As EV charging systems become increasingly connected, secure communication protocols and robust data management frameworks are essential. Future work should address cybersecurity risks in charging stations, distributed energy resources, and V2G systems. Data-driven control strategies leveraging telematics and real-time monitoring will be critical for adaptive infrastructure management.

While theoretical optimization models provide valuable insights, practical deployment of EV charging infrastructure involves several engineering and regulatory constraints that influence feasibility and scalability. One major challenge lies in grid connection approval procedures. Installation of medium- and high-power charging stations typically requires utility clearance, transformer capacity assessment, feeder reinforcement studies, and compliance with voltage stability and protection standards. In many regions, grid interconnection processes can extend project timelines due to load impact analysis and distribution network upgrade requirements. Consequently, optimization models must account not only for ideal load allocation but also for existing feeder capacity, transformer loading margins, and fault-level constraints.

Land-use and spatial planning considerations also play a critical role. Urban charging stations require zoning approvals, parking space allocation, accessibility compliance, and integration with existing transportation infrastructure. Highway fast-charging hubs demand strategic placement near service corridors, adequate land footprint for multiple charging bays, and access to high-capacity distribution lines. Physical space limitations, especially in dense metropolitan areas, often constrain optimal station placement identified by purely mathematical models.

Economic feasibility further affects implementation. High-power DC fast chargers involve substantial capital expenditure, including grid upgrades, protection equipment, power electronics, civil works, and communication infrastructure. Investment recovery depends on utilization rate, electricity tariff structure, demand charges, and EV penetration levels. In early deployment stages, low utilization rates may extend payback periods, influencing investor decisions and infrastructure expansion strategies.

Finally, integration with transportation systems introduces operational challenges such as traffic congestion at charging hubs, queue spillover into public roads, and coordination with traffic management authorities. Engineering solutions must therefore consider not only electrical performance but also traffic flow regulation and user accessibility. Addressing these practical constraints is essential for translating optimization frameworks into deployable infrastructure strategies. Future research should therefore integrate technical optimization with regulatory, spatial, and techno-economic considerations to ensure realistic and scalable implementation pathways.

7 Conclusions

While EV technologies have advanced significantly in recent years, several limitations remain in both research and practical implementation. Much of the existing literature focuses on technological improvements and algorithmic development, while comparatively less attention has been given to systematic evaluation of real-world infrastructure constraints, economic feasibility, and large-scale deployment challenges. In addition, many optimization approaches are primarily validated through simulations, with limited demonstration under realistic transportation and grid operating conditions.

This review has examined recent developments in EV technologies, charging methods, and optimization strategies, emphasizing their potential role in enabling sustainable transportation systems. The rapid growth of EV adoption has intensified the need for effective solutions addressing battery performance, grid integration, and charging infrastructure deployment. A variety of charging approaches, including conventional CC, wireless charging, and

dynamic in-motion charging, have been investigated to support growing demand. However, further improvements in efficiency, charging speed, and infrastructure scalability are still required to support widespread deployment. Optimization techniques have also been widely explored to reduce charging costs, minimize power losses, and enhance grid stability under increasing EV penetration.

Future work should focus on developing traffic-integrated charging infrastructure planning models, scalable real-time optimization algorithms, and robust coordination frameworks between EV fleets and power distribution networks. Progress in battery technology, smart charging systems, and V2G integration will also play an important role in improving system efficiency and operational flexibility. Additionally, the integration of EVs with emerging mobility paradigms such as autonomous vehicles and Mobility-as-a-Service (MaaS) will require new approaches to fleet management and energy coordination. Furthermore, issues related to cybersecurity, data privacy, and lifecycle environmental impacts must be carefully addressed to ensure reliable and sustainable EV deployment. Comprehensive lifecycle assessment and sustainable battery manufacturing practices will be necessary to fully realize the environmental benefits of electrified transportation. In summary, while considerable progress has been made in EV technologies and optimization frameworks, achieving large-scale sustainable deployment will require continued interdisciplinary research that integrates technological innovation, infrastructure planning, and practical implementation considerations within modern transportation systems.

Author Contributions

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors utilized ChatGPT to refine the language. Afterward, they reviewed and edited the content as necessary and took full responsibility for the publication's content.

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